LAB 9 : Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

OBJECTIVES :

- To configure a wireless router and establish a wireless network using Cisco Packet Tracer.
- To connect wireless devices to the network and assign IP addresses using DHCP.
- To verify secure wireless communication using WPA2 security settings.

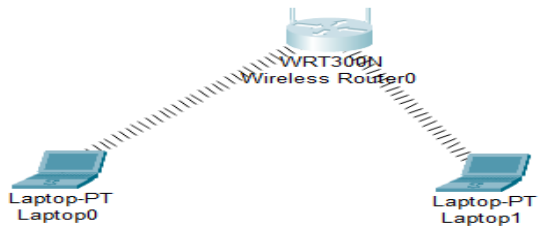
THEORY :

A wireless network enables devices to communicate without physical connections by using radio waves. In this lab, a wireless router was configured to establish a wireless local area network (WLAN), providing connectivity, automatic IP address assignment via DHCP, and secure access for wireless devices. The router assigns IP addresses to connected clients, facilitating efficient network communication.

The network is identified by an SSID, while WPA2 security with AES encryption ensures that communication remains protected from unauthorized access. Wireless devices connect using the correct SSID and passphrase, and connectivity is verified through network testing.

Packet analysis is employed to monitor and understand data flow between the router and wireless devices. Using Cisco Packet Tracer's simulation and packet inspection tools, various packet types—including DHCP, ARP, and data packets—can be examined. This analysis demonstrates how devices acquire IP addresses, establish connections, and transmit data over a wireless network. It also aids in verifying successful communication and troubleshooting, making packet analysis an essential component of wireless network implementation.

Network Topology :



Procedure :

Wireless Router Configuration :

Step 1 : Basic Setup

1. Click on the Wireless Router.
2. Go to GUI → Setup.
3. Configure the following:
   - IP Address: 192.168.0.1
   - Subnet Mask: 255.255.255.0
   - DHCP Server: Enabled

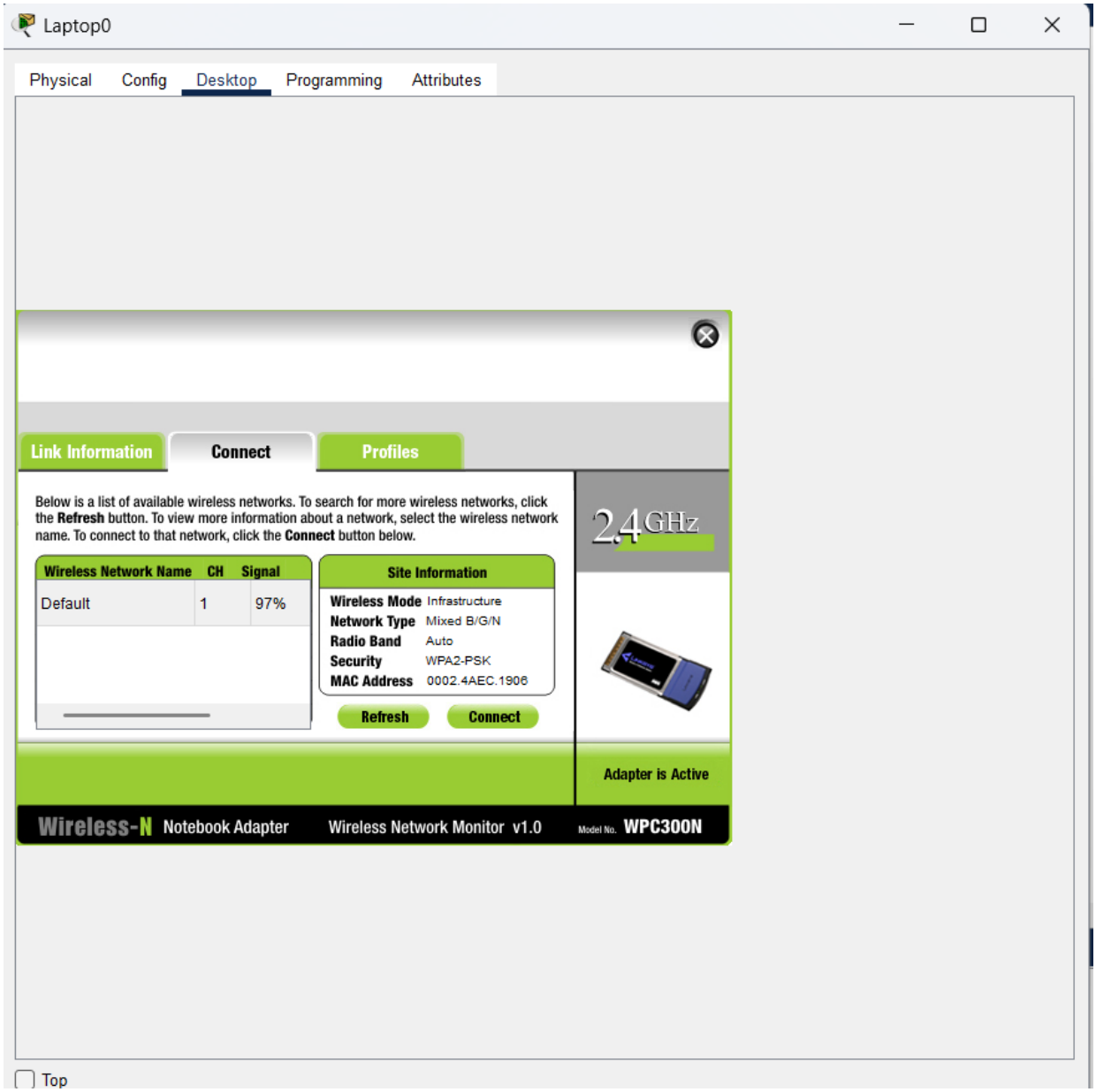
Step 2 : Wireless Configuration

1. Go to Wireless → Basic Wireless Settings.
2. Configure the following parameters:
3. Network Mode: Mixed
   - SSID: HCOE
   - Channel: Auto
4. Click Save Settings.

Step 3 : Wireless Security

1. Go to Wireless → Wireless Security.
2. Configure the following:
   - Security Mode: WPA2-Personal
   - Encryption: AES • Passphrase: hcoe@123
3. Save the configuration.

Observation :



RESULT :

The setup successfully established a functional wireless network using a WRT300N router. The SSID was set to "HCOE," and the DHCP server was enabled with the gateway address 192.168.0.1, preparing the network for client connections. Both Laptop0 and Laptop1 detected the network at full signal strength and joined successfully after authenticating with the WPA2PSK passphrase "hcoe@123." As a result, both devices received IP addresses automatically and were able to exchange data packets through the router, confirming proper network operation.

DISCUSSION :

This lab demonstrates how DHCP and WPA2 security combine to establish a functional and secure wireless network. Enabling the DHCP server allowed the router to automatically assign IP addresses to both laptops, providing a faster and more reliable alternative to manual configuration. Configuring WPA2-Personal with AES encryption was crucial, as it safeguards the network by requiring a specific passphrase to prevent unauthorized access.

CONCLUSION :

The lab successfully demonstrated that a secure, functional wireless network can be established by correctly configuring SSID, WPA2 security, and DHCP services.